Enterprise LAN Design and Security Assessment Report

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Course: IT 335 / Corporate Cybersecurity
Assignment: Enterprise LAN Design and Security Assessment

Objective

The purpose of this lab was to design and configure a small enterprise network in Cisco Packet Tracer. The network included three separate LANs connected through routers. Static routing was configured so that all networks could communicate with each other successfully.

The lab also included testing web access using HTTP and HTTPS, verifying switch MAC address learning, and troubleshooting routing problems.

Network Topology

Insert your topology screenshot here.

The network contains:

- Three routers
- Three switches
- Multiple PCs
- One server
- One attacker PC
- Three separate LANs

IP ranges used:

- IT Network: 10.1.10.0/24
- Server Network: 192.168.50.0/24
- Remote Network: 172.16.5.0/24

Router 1 Configuration

Show IP Interface Brief

Insert Router1 interface screenshot here.

Router1 interfaces:

- GigabitEthernet0/0 = 10.1.10.1
- GigabitEthernet0/1 = 10.255.1.1

All interfaces were up and operational.

Show IP Route

Insert Router1 routing table screenshot here.

Router1 had static routes configured to reach:

- 172.16.5.0/24
- 192.168.50.0/24

Router 0 Configuration

Show IP Interface Brief

Insert Router0 interface screenshot here.

Router0 interfaces:

- GigabitEthernet0/0 = 192.168.50.1
- GigabitEthernet0/1 = 10.255.1.2
- GigabitEthernet0/2 = 10.255.2.1

All interfaces were up and operational.

Show IP Route

Insert Router0 routing table screenshot here.

Router0 had static routes configured to reach:

- 10.1.10.0/24

- 172.16.5.0/24

Router 2 Configuration

Show IP Interface Brief

Insert Router2 interface screenshot here.

Router2 interfaces:

- GigabitEthernet0/0 = 172.16.5.1
- GigabitEthernet0/1 = 10.255.2.2

All interfaces were up and operational.

Show IP Route

Insert Router2 routing table screenshot here.

Router2 had static routes configured to reach:

- 10.1.10.0/24
- 192.168.50.0/24

Switch MAC Address Table

Insert switch MAC address table screenshot here.

The switch successfully learned MAC addresses dynamically from connected devices. This confirms that the switch was functioning correctly at Layer 2.

Connectivity Testing

Insert ping test screenshot here.

Ping tests were performed between different networks.

Successful tests included:

- Ping from remote network to 172.16.5.1

- Ping from remote network to 10.1.10.10
- Ping from remote network to 192.168.50.10

All pings were successful with 0% packet loss.

This confirmed that static routing between all routers was working correctly.

HTTP and HTTPS Testing

HTTP Test

Insert HTTP screenshot here.

The server was configured with HTTP enabled. A PC successfully opened the server web page using:

`http://192.168.50.10`

HTTPS Test

Insert HTTPS screenshot here.

The server was also configured with HTTPS enabled. A PC successfully opened the secure web page using:

`https://192.168.50.10`

This demonstrated secure web communication in the network.

Troubleshooting

During the lab, some devices could not initially communicate with other networks.

The following troubleshooting steps were used:

1. Verified IP addresses and subnet masks on all devices
2. Verified default gateways on PCs and server
3. Used `show ip interface brief` to confirm interfaces were up
4. Used `show ip route` to verify routing tables
5. Added missing static routes where necessary
6. Retested connectivity with ping commands

After troubleshooting, all devices were able to communicate successfully.

Conclusion

This lab successfully demonstrated how to build and secure a small enterprise network using Cisco Packet Tracer.

The routers were configured with static routes, switches learned MAC addresses correctly, and all PCs were able to communicate across networks.

HTTP and HTTPS services were also tested successfully.

Overall, the lab showed the importance of proper IP addressing, routing, troubleshooting, and secure communication in enterprise networks.

```
Router>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.1.10.1       YES manual up                    up
GigabitEthernet0/1     10.255.1.1      YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router>
```

```
Router>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.50.1    YES manual up                    up
GigabitEthernet0/1     10.255.1.2      YES manual up                    up
GigabitEthernet0/2     10.255.2.1      YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Router>
```

```
Router>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.50.1    YES manual up                    up
GigabitEthernet0/1     10.255.1.2      YES manual up                    up
GigabitEthernet0/2     10.255.2.1      YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 5 subnets, 3 masks
S       10.1.10.0/24 [1/0] via 10.255.1.1
C       10.255.1.0/30 is directly connected, GigabitEthernet0/1
L       10.255.1.2/32 is directly connected, GigabitEthernet0/1
C       10.255.2.0/30 is directly connected, GigabitEthernet0/2
L       10.255.2.1/32 is directly connected, GigabitEthernet0/2
     172.16.0.0/24 is subnetted, 1 subnets
S       172.16.5.0/24 [1/0] via 10.255.2.2
     192.168.50.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.50.0/24 is directly connected, GigabitEthernet0/0
L       192.168.50.1/32 is directly connected, GigabitEthernet0/0

Router>
```

Copy

```
Router>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     172.16.5.1      YES manual up                    up
GigabitEthernet0/1     10.255.2.2      YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router>
```

```
Router>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     172.16.5.1      YES manual up                    up
GigabitEthernet0/1     10.255.2.2      YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 3 subnets, 3 masks
S       10.1.10.0/24 [1/0] via 10.255.2.1
C       10.255.2.0/30 is directly connected, GigabitEthernet0/1
L       10.255.2.2/32 is directly connected, GigabitEthernet0/1
     172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.5.0/24 is directly connected, GigabitEthernet0/0
L       172.16.5.1/32 is directly connected, GigabitEthernet0/0
S    192.168.50.0/24 [1/0] via 10.255.2.1

Router>
```

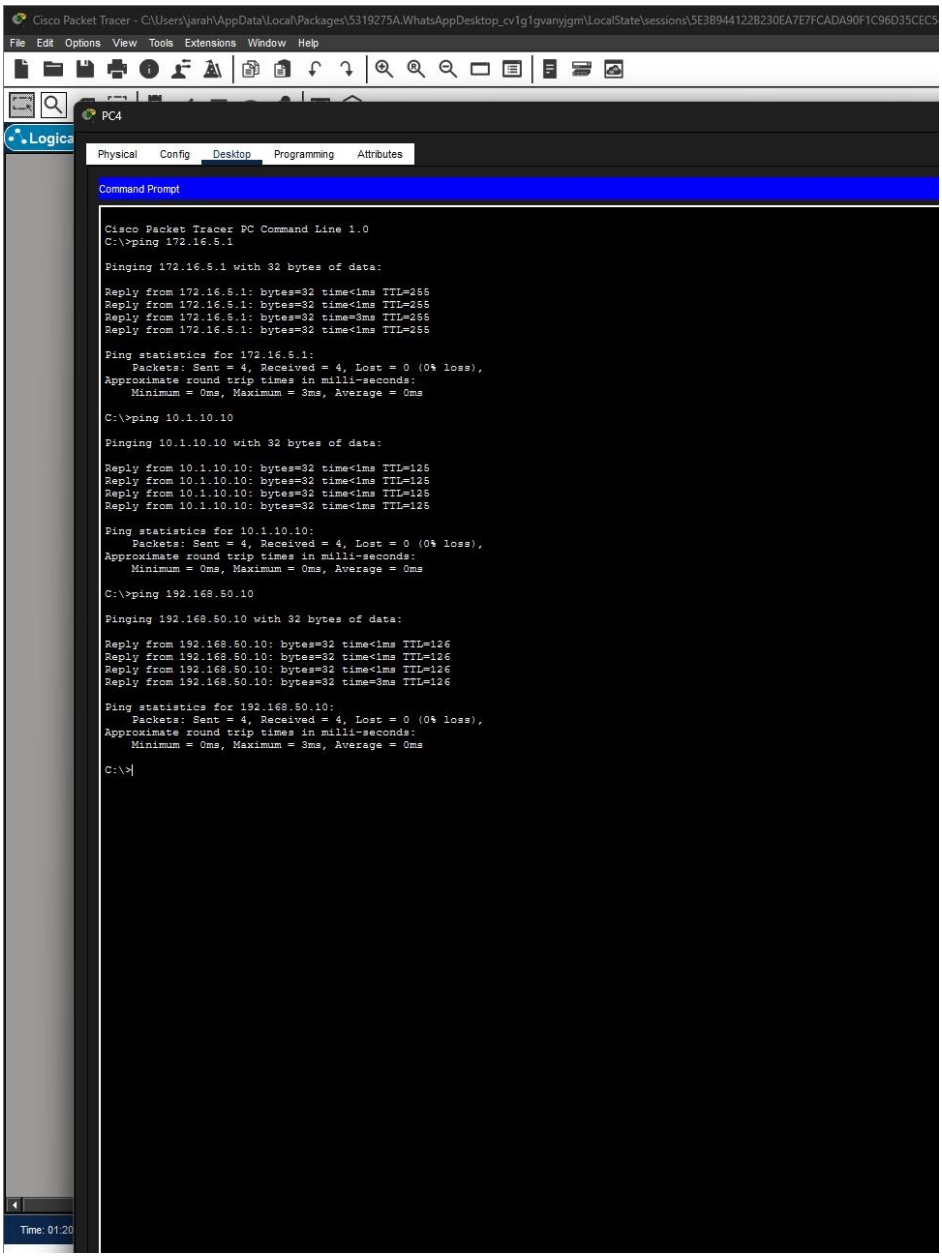
```
Router>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.1.10.1       YES manual up                    up
GigabitEthernet0/1     10.255.1.1      YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
C       10.1.10.0/24 is directly connected, GigabitEthernet0/0
L       10.1.10.1/32 is directly connected, GigabitEthernet0/0
C       10.255.1.0/30 is directly connected, GigabitEthernet0/1
L       10.255.1.1/32 is directly connected, GigabitEthernet0/1
     172.16.0.0/24 is subnetted, 1 subnets
S       172.16.5.0/24 [1/0] via 10.255.1.2
S    192.168.50.0/24 [1/0] via 10.255.1.2

Router>
```

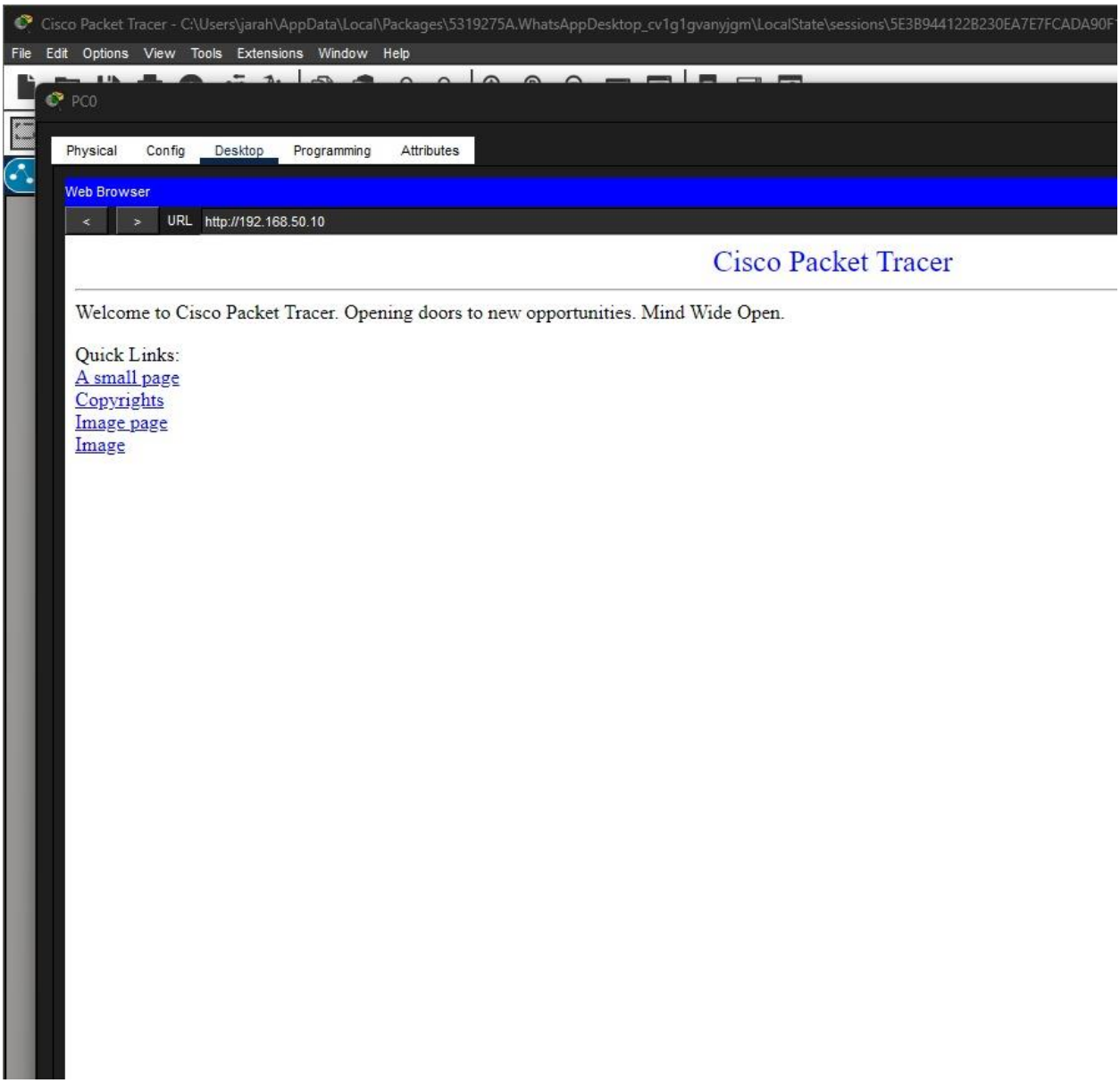
PC0

Physical | Config | Desktop | Programming | Attributes

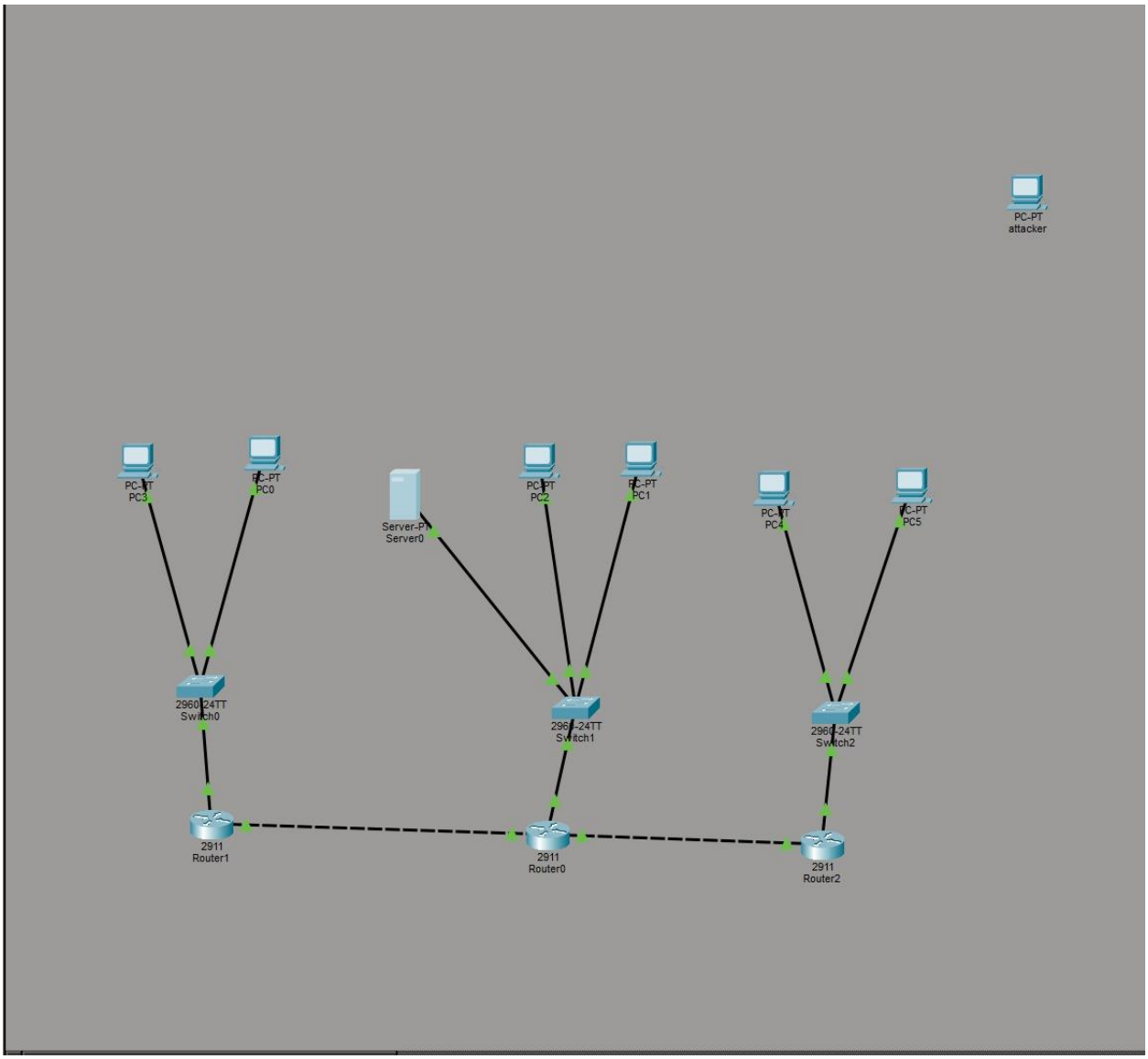
Command Prompt

```
C:\>ping 192.168.50.10

Pinging 192.168.50.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.50.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 172.16.5.1

Pinging 172.16.5.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 172.16.5.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 172.16.5.10

Pinging 172.16.5.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 172.16.5.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.50.10

Pinging 192.168.50.10 with 32 bytes of data:

Reply from 192.168.50.10: bytes=32 time<1ms TTL=126
Reply from 192.168.50.10: bytes=32 time<1ms TTL=126
Reply from 192.168.50.10: bytes=32 time<1ms TTL=126
Reply from 192.168.50.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.50.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 172.16.5.10

Pinging 172.16.5.10 with 32 bytes of data:

Reply from 172.16.5.10: bytes=32 time<1ms TTL=125
Reply from 172.16.5.10: bytes=32 time<1ms TTL=125
Reply from 172.16.5.10: bytes=32 time<1ms TTL=125
Reply from 172.16.5.10: bytes=32 time<1ms TTL=125

Ping statistics for 172.16.5.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.50.10

Pinging 192.168.50.10 with 32 bytes of data:

Reply from 192.168.50.10: bytes=32 time<1ms TTL=126
Reply from 192.168.50.10: bytes=32 time<1ms TTL=126
Reply from 192.168.50.10: bytes=32 time<1ms TTL=126
Reply from 192.168.50.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.50.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 172.16.5.10

Pinging 172.16.5.10 with 32 bytes of data:

Reply from 172.16.5.10: bytes=32 time=2ms TTL=125
Reply from 172.16.5.10: bytes=32 time<1ms TTL=125
Reply from 172.16.5.10: bytes=32 time=1ms TTL=125
Reply from 172.16.5.10: bytes=32 time<1ms TTL=125

Ping statistics for 172.16.5.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>
```


Cisco Packet Tracer - C:\Users\jarah\AppData\Local\Packages\5319275A.WhatsAppDesktop_cv1g1gvanyjgm\LocalState\sessions\5E3B9441228230EA7E7FCADA90F1C96D35CEC5
File Edit Options View Tools Extensions Window Help
PC4
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.5.1
Pinging 172.16.5.1 with 32 bytes of data:
Reply from 172.16.5.1: bytes=32 time<1ms TTL=255
Reply from 172.16.5.1: bytes=32 time<1ms TTL=255
Reply from 172.16.5.1: bytes=32 time=3ms TTL=255
Reply from 172.16.5.1: bytes=32 time<1ms TTL=255
Ping statistics for 172.16.5.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 3ms, Average = 0ms
C:\>ping 10.1.10.10
Pinging 10.1.10.10 with 32 bytes of data:
Reply from 10.1.10.10: bytes=32 time<1ms TTL=125
Reply from 10.1.10.10: bytes=32 time<1ms TTL=125
Reply from 10.1.10.10: bytes=32 time<1ms TTL=125
Reply from 10.1.10.10: bytes=32 time<1ms TTL=125
Ping statistics for 10.1.10.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.50.10
Pinging 192.168.50.10 with 32 bytes of data:
Reply from 192.168.50.10: bytes=32 time<1ms TTL=126
Reply from 192.168.50.10: bytes=32 time<1ms TTL=126
Reply from 192.168.50.10: bytes=32 time<1ms TTL=126
Reply from 192.168.50.10: bytes=32 time=3ms TTL=126
Ping statistics for 192.168.50.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 3ms, Average = 0ms
C:\>
Time: 01:20


Cisco Packet Tracer - C:\Users\jarah\AppData\Local\Packages\5319275A.WhatsAppDesktop_cv1g1gvanyjgm\LocalState\sessions\5E3B944122B230EA7E7FCADA90F
File Edit Options View Tools Extensions Window Help
PC0
Physical Config Desktop Programming Attributes
Web Browser
< > URL http://192.168.50.10
Cisco Packet Tracer
Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.
Quick Links:
A small page
Copyrights
Image page
Image

PC0
Physical
Config
Desktop
Programming
Attributes
Web Browser
<
>
URL
https://192.168.50.10
Cisco Packet Tracer
Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.
Quick Links:
A small page
Copyrights
Image page
Image


PC-PT
attacker
PC-PT
PC3
PC-PT
PC0
Server-PT
Server0
PC-PT
PC2
PC-PT
PC1
PC-PT
PC4
PC-PT
PC5
2960-24TT
Switch0
2960-24TT
Switch1
2960-24TT
Switch2
2911
Router1
2911
Router0
2911
Router2